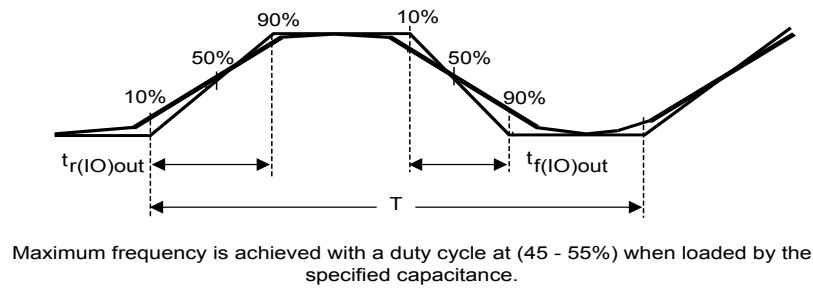


**Figure 21. Output AC characteristics definition**


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### 5.3.15 NRST pin characteristics

The NRST pin input driver uses the CMOS technology. It is connected to a permanent pullup resistor,  $R_{PU}$ .

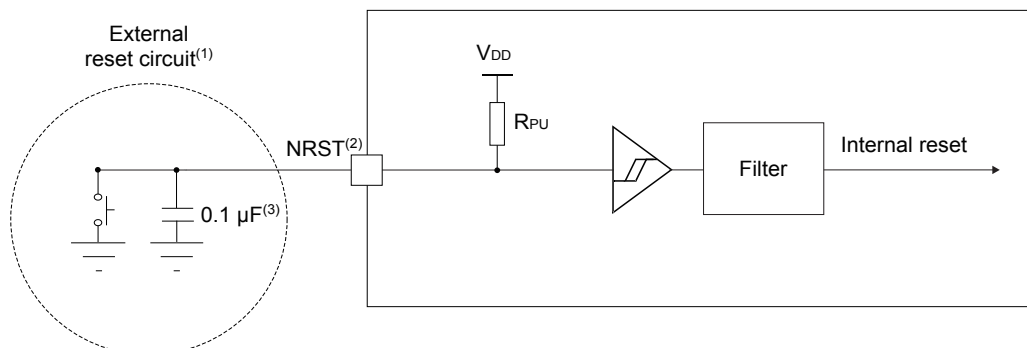
Unless otherwise specified, the parameters given in the table below are derived from tests performed under the ambient temperature and supply voltage conditions summarized in Table 18.

**Table 52. NRST pin characteristics**

Specified by design and not tested in production.

| Symbol          | Parameter                                       | Conditions                     | Min                    | Typ | Max                    | Unit       |
|-----------------|-------------------------------------------------|--------------------------------|------------------------|-----|------------------------|------------|
| $V_{IL(NRST)}$  | NRST input low-level voltage                    | -                              | -                      | -   | $0.3 \times V_{DDIOx}$ | V          |
| $V_{IH(NRST)}$  | NRST input high-level voltage                   | -                              | $0.7 \times V_{DDIOx}$ | -   | -                      |            |
| $V_{hys(NRST)}$ | NRST Schmitt trigger voltage hysteresis         | -                              | -                      | 200 | -                      | mV         |
| $R_{PU}$        | Weak pull-up equivalent resistor <sup>(1)</sup> | $V_{IN} = V_{SS}$              | 30                     | 40  | 50                     | k $\Omega$ |
| $t_{F(NRST)}$   | NRST input filtered pulse                       | -                              | -                      | -   | 50                     | ns         |
| $t_{NF(NRST)}$  | NRST input not-filtered pulse                   | $2.7 V \leq V_{DD} \leq 3.6 V$ | 350                    | -   | -                      |            |

1. The pull-up is designed with a true resistance in series with a switchable PMOS. This PMOS contribution to the series resistance is minimal (~10 % order).

**Figure 22. Recommended NRST pin protection**


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- (1): The reset network protects the device against parasitic resets.
- (2): The user must ensure that the level on the NRST pin can go below the  $V_{IL(NRST)}$  max level specified in the above table. Otherwise the reset is not taken into account by the device.
- (3): The external capacitor on NRST must be placed as close as possible to the device.